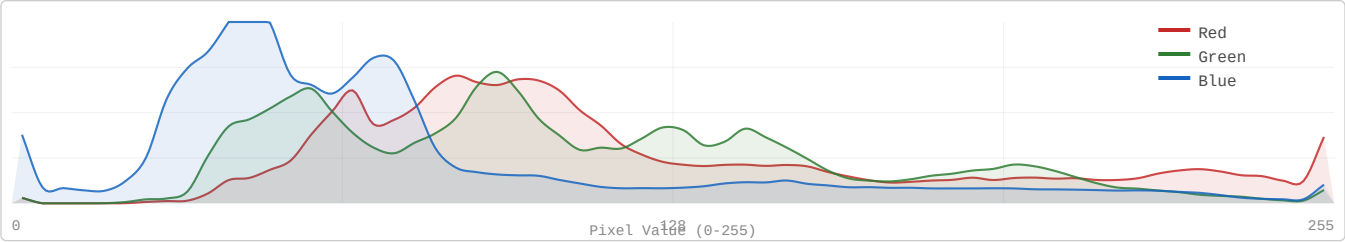
Decorrelation Gap: KODIM23

768x512 | 24-bit RGB | PNG (lossless)

Parrot Close-up | Three-dimensional (PC1 = 70.3%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	**1.0000**	0.6147	0.4231
G	0.6147	**1.0000**	0.6408
B	0.4231	0.6408	**1.0000**

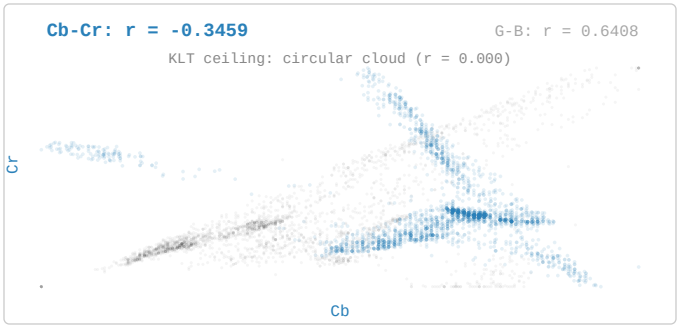
Cb-Cr: -0.346

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	**1.0000**	-0.2343	0.0095
Cb	-0.2343	**1.0000**	-0.3459
Cr	0.0095	-0.3459	**1.0000**

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.614747	-0.234260	0.2343
R-B → Y-Cr	0.423120	0.009543	0.0095
G-B → Cb-Cr	0.640769	-0.345934	0.3459
Avg |r|	0.559545	0.196579	0.1966

Decorrelation Gap: KODIM23

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Parrot Close-up | Three-dimensional (PC1 = 70.3%)

5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

64.9% High

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

35.9%

0% Fully predictable — 100% Fully independent

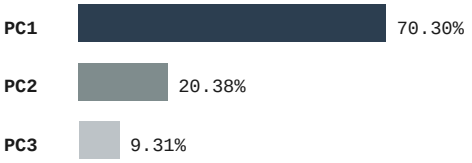
6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	42.40 dB
G	48.10 dB
B	39.97 dB
Overall	42.37 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.6100	-0.5732	-0.5472	6155.85
PC2	-0.7170	0.1051	0.6891	1784.85
PC3	0.3375	-0.8126	0.4751	815.24

8. PCA Derived Metrics

Condition Number

7.55

PC1/PC2: 3.45 | PC2/PC3: 2.19

Blue Channel Independence

35.9%

Variance not captured by PC1

PC1 Dominant Channel

R

Loading: R (0.61)

9. Redundancy Profile

Classification: Three-dimensional | CN: 7.55 | PC1: 70.3%

RGB Correlation: Avg |r| = 0.559545 | R-G: 0.6147 | R-B: 0.4231 | G-B: 0.6408

YCbCr Residual: Avg |r| = 0.196579 | Y-Cb: -0.2343 | Y-Cr: 0.0095 | Cb-Cr: -0.3459

Efficiency: 64.9% of KLT ceiling | Gap: 0.1966 avg |r| remaining

Blue Channel: 35.9% independent | 4:2:0 PSNR: 42.37 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.